UNIFIED MODELING LANGUAGE (UML)

Introduction to UML

- UML stands for Unified Modeling Language
- It is a standard graphical language used to visualize software systems
- Developed by Grady Booch, James Rumbaugh, and Ivar Jacobson
- Used to design, model, and document system architecture
- Supports object-oriented concepts like classes and objects

Why UML is Important

- Helps in better understanding of complex systems
- Provides a visual representation of system design
- Improves communication between team members
- Helps in early detection of errors and issues

Types of UML Diagrams

- Acts as a blueprint before coding begins
- Reduces development time and cost

Types of UML Diagrams

•UML diagrams are categorized into:

•Structural Diagrams (Static View)

•Behavioral Diagrams (Dynamic View)

•Structural diagrams show system components

•Behavioral diagrams show system behavior over time

•Each diagram focuses on a different perspective

Structural Diagrams

•Represent the static structure of a system

•Show classes, objects, and relationships

Important types:

•Class Diagram (most commonly used)

•Object Diagram

•Component Diagram

•Deployment Diagram

•Used during design phase

Behavioral Diagrams

•Represent the dynamic aspects of a system

•Show how system behaves during execution

•Important types:

•Use Case Diagram

•Sequence Diagram

•Activity Diagram

•State Diagram

Class Diagram

•Describes the structure of the system

•Contains:

•Class name

•Attributes (data members)

•Methods (function

•Shows relationships:

•Inheritance (IS-A)

•Association (uses)

•Aggregation (has-a)

Use Case Diagram

•Represents functional requirements of the system

•Shows interaction between:

•Actors (users)

•System (use cases)

•Helps in requirement analysis

•Includes relationships:

•Include

•Extend

•Easy to understand for both technical & non-technical users

Sequence & Activity Diagrams

- Sequence Diagram:
- Shows interaction between objects in time order
- Displays messages exchanged
- Helps understand system flow step-by-step
- Activity Diagram:
- Represents workflow of activities
- Includes:
  - *Start/End points*
  - *Decisions*
  - *Loops*
- Useful for business process modeling

THANK YOU